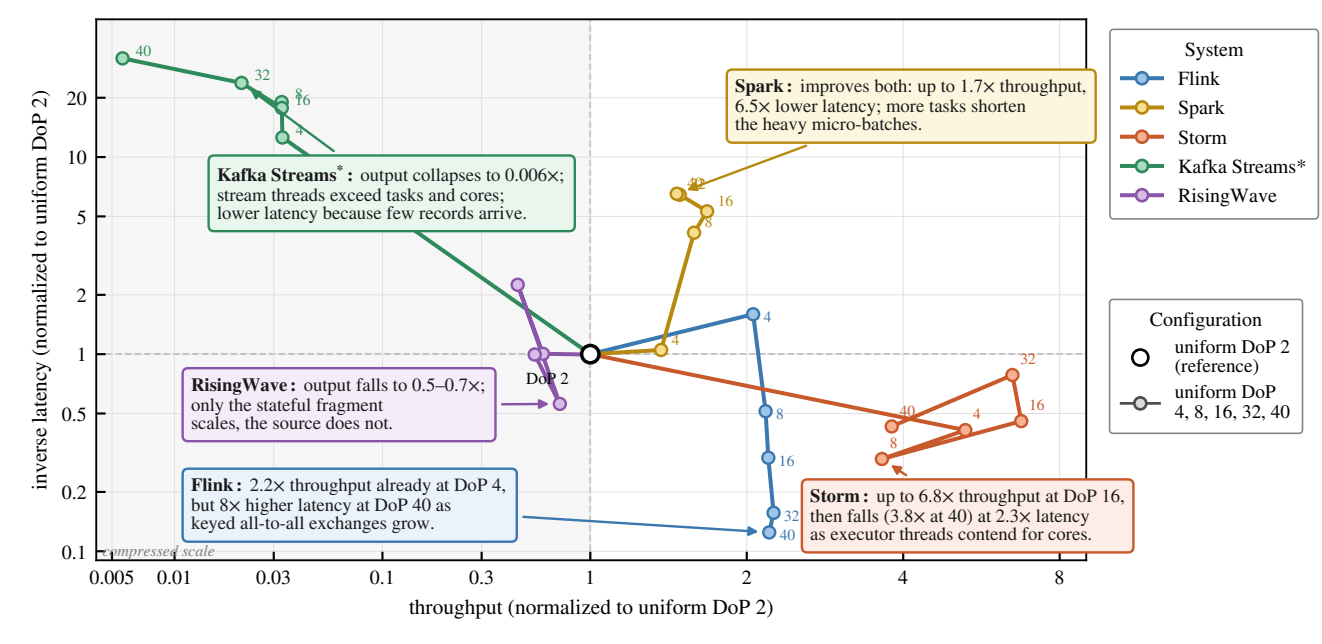
System
Flink
Spark
Storm
Kafka Streams*
RisingWave
Configuration
uniform DoP 2 (reference)
uniform DoP 4, 8, 16, 32, 40
Spark: improves both: up to 1.7× throughput, 6.5× lower latency; more tasks shorten the heavy micro-batches.
Kafka Streams*: output collapses to 0.006×; stream threads exceed tasks and cores; lower latency because few records arrive.
RisingWave: output falls to 0.5–0.7×; only the stateful fragment scales, the source does not.
Flink: 2.2× throughput already at DoP 4, but 8× higher latency at DoP 40 as keyed all-to-all exchanges grow.
Storm: up to 6.8× throughput at DoP 16, then falls (3.8× at 40) at 2.3× latency as executor threads contend for cores.
DoP 2
compressed scale
inverse latency (normalized to uniform DoP 2)
throughput (normalized to uniform DoP 2)
0.005
0.01
0.03
0.1
0.3
1
2
4
8
0.1
0.2
0.5
1
2
5
10
20